Answer **all** the questions in the spaces provided.

1 **Determination of a value for an enthalpy change of decomposition, ΔH^1_{decomp}**

Hydrogen peroxide decomposes very slowly to form water and oxygen gas as shown in equation **1**. The enthalpy change of decomposition, ΔH_{decomp}, for this reaction is exothermic.

equation **1** $2H_2O_2 \rightarrow 2H_2O + O_2$ ΔH_{decomp}

Many transition element ions are able to catalyse the decomposition of hydrogen peroxide. Iron(III) nitrate, $Fe(NO_3)_3$, is an effective catalyst for this reaction.

FA 1 is 0.850 mol dm^{-3} aqueous hydrogen peroxide, H_2O_2.
FA 2 is iron(III) nitrate, $Fe(NO_3)_3$.

The addition of **FA 2** to **FA 1** increases the rate of decomposition of H_2O_2. This increase is sufficient to allow the maximum temperature change, ΔT_{max}, to be determined within a few minutes.

In this question, you will perform an experiment to determine a value for ΔT_{max}. You will then repeat this experiment and hence determine the average maximum temperature change, ΔT^1_{max}.

You will use your value of ΔT^1_{max} to calculate a value for the enthalpy change for the decomposition reaction shown in equation **1**, ΔH^1_{decomp}.

In **1(d)**, you will use data provided to determine another value for the maximum temperature change, ΔT^2_{max}. You will then calculate a second value for the enthalpy change for the decomposition reaction shown in equation **1**, ΔH^2_{decomp}.

(a) Perform the tests described in Table 1.1, and record your observations in the table. Test and identify any gases evolved.

Table 1.1

	tests	observations
1.	Test the **FA 2** solution using Universal Indicator paper.	Indicator paper turns red (pH ~2) or orange (pH ~3)
2.	Put about a 2 cm depth of **FA 1** into a test-tube. Add about a 2 cm depth of **FA 2** to the same test-tube and shake the mixture thoroughly. Observe the mixture until no further changes are seen.	Effervescence observed Colourless, odourless gas relights glowing splint. Hence, oxygen gas is present Orange solution turns brown then yellow
While you are waiting, continue with (b).		

[2]

(b) Determination of the average maximum temperature change, ΔT^1_{max}, for the catalysed decomposition of hydrogen peroxide, FA 1

You will follow the instructions to perform the experiment twice, run A and run B.

Record your results in Table 1.2.

(i) 1. Using a measuring cylinder, transfer 50.0 cm^3 of **FA 1** into a Styrofoam cup. Place this cup inside a second Styrofoam cup, which is placed in a 250 cm^3 glass beaker.
2. Stir and measure the temperature of this **FA 1**, $T_{\mathbf{FA\,1}}$.
3. Using another measuring cylinder, measure 50.0 cm^3 of **FA 2**.
4. Stir and measure the temperature of this **FA 2**, $T_{\mathbf{FA\,2}}$.
5. Add the **FA 2** from the measuring cylinder to the **FA 1** in the Styrofoam cup.
6. Using the thermometer, stir the mixture continuously until it reaches its maximum temperature. This could take several minutes. Record this temperature, T_{max}.
7. Replace the wet Styrofoam cup with a clean, dry Styrofoam cup. Repeat steps **1** to **6** to obtain a second value for T_{max}.

Complete Table 1.2 by calculating values for the average of $T_{\mathbf{FA\,1}}$ and $T_{\mathbf{FA\,2}}$, T_{ave}, and ΔT_{max} for each run.

Table 1.2

	run A	run B
$T_{\mathbf{FA\,1}}$ / °C	29.6	29.4
$T_{\mathbf{FA\,2}}$ / °C	29.6	29.4
T_{ave} / °C	29.6	29.4
T_{max} / °C	39.6	39.6
ΔT_{max} / °C	10.0	10.2

Determine the average maximum temperature change, ΔT^1_{max}.

Average maximum temperature change, $\Delta T^1_{max} = \dfrac{10.0 + 10.2}{2} =$ **10.1 °C**

ΔT^1_{max} = ...10.1.......... °C [2]

(ii) Calculate the average heat change, q, for your experiment in **1(b)(i)** and hence determine a value for the enthalpy change, for the decomposition reaction shown in equation **1**, ΔH^1_{decomp}.

You should assume that the:

- decomposition of H_2O_2 in **FA 1** is complete when T_{max} is reached;
- specific heat capacity of the mixture is $4.18\,J\,g^{-1}\,K^{-1}$;
- density of the mixture is $1.00\,g\,cm^{-3}$.

Average heat change, $q = -mc\Delta T^1_{max} = -(100 \times 1.00) \times 4.18 \times 10.1$

$\mathbf{= -4220\ J = -4.22\ kJ}$

q = ...–4.22 kJ....................

Amount of H_2O_2 used $= 0.850 \times \dfrac{50}{1000} = \mathbf{0.0425\ mol}$

$\Delta H = \dfrac{-4.22\ kJ}{0.0425\ mol} = -99.3\ kJ\ mol^{-1}$ $\left(\text{for } H_2O_2 \rightarrow H_2O + \tfrac{1}{2}O_2\right)$

Equation 1: $2H_2O_2 \rightarrow 2H_2O + O_2$ $\left(i.e.\ \Delta H^1_{decomp} = 2\Delta H\right)$; $\therefore \Delta H^1_{decomp} = 2 \times -99.3 = \underline{\mathbf{-199\ kJ\ mol^{-1}}}$

ΔH^1_{decomp} = ...–199 kJ mol⁻¹.................... [3]

(c) Steps **1** to **4** represent a possible mechanism for the catalysed decomposition of H_2O_2. In this mechanism, the O_2H ligand on one of the complex ions represents the H–O–O⁻ ion and the O represents an oxygen atom.

step **1** $\quad [Fe(H_2O)_6]^{3+}(aq) + H_2O_2(aq) \rightleftharpoons [Fe(H_2O)_5(O_2H)]^{2+}(aq) + H_3O^+(aq)$

step **2** $\quad [Fe(H_2O)_5(O_2H)]^{2+}(aq) \rightarrow OH^-(aq) + [Fe(H_2O)_5O]^{3+}(aq)$

step **3** $\quad [Fe(H_2O)_5O]^{3+}(aq) + H_2O_2(aq) \rightarrow [Fe(H_2O)_6]^{3+}(aq) + O_2(g)$

step **4** $\quad H_3O^+(aq) + OH^-(aq) \rightarrow 2H_2O(l)$

(i) Explain fully how you can tell that the $[Fe(H_2O)_6]^{3+}$ ions are acting as a catalyst in this reaction, using evidence from your observations in Table 1.1 and from the mechanism.

Catalyst is a substance that increases the rate of reaction by providing an alternate reaction pathway of lower activation energy, without itself undergoing any permanent chemical change.

From the **mechanism**, **$[Fe(H_2O)_6]^{3+}$ reacted with H_2O_2 in step 1 and was regenerated in step 3**, implying that it is a catalyst.

From the **observation**, **orange (yellow) $[Fe(H_2O)_6]^{3+}$ reacted with H_2O_2 changes to brown and then back to yellow again**, implying that **$[Fe(H_2O)_6]^{3+}$ is regenerated** and it is a catalyst.

Compared to an absence of $[Fe(H_2O)_6]^{3+}$, **oxygen gas was produced rapidly and vigorously** when $[Fe(H_2O)_6]^{3+}$ was added, implying that the **rate increases**.

.... [3]

(ii) The crystals of hydrated iron(III) nitrate, $Fe(NO_3)_3.9H_2O$, used to prepare your sample of **FA 2** were pale violet in colour.

The colour of the **FA 2** solution obtained by dissolving these crystals is due to the $[Fe(H_2O)_6]^{3+}$ ion reacting with a water molecule.

Complete the equation and explain how you deduced the products of this reaction, using your observations from test 1 in Table 1.1 to help you.

$[Fe(H_2O)_6]^{3+} + H_2O \rightarrow$ $[Fe(H_2O)_5(OH)]^{2+}$ + H_3O^+

explanation **The Universal Indicator turned red, implying that it is acidic (pH ~ 2)**. This indicates the presence of H_3O^+ ions. Due to **high charge density of Fe^{3+}**, **$[Fe(H_2O)_6]^{3+}$ hydrolyses in water to produce an acidic solution** as the O–H bond in H_2O is weakened and broken due to the high polarising power of Fe^{3+}.

[2]

(iii) Explain the colour changes you observed for test 2 in Table 1.1, in terms of the complex ions involved in the mechanism.

In test 2, there are **ligand exchanges** taking place in the complex ions involved, from **$[Fe(H_2O)_6]^{3+}$ to $[Fe(H_2O)_5(O_2H)]^{2+}$ to $[Fe(H_2O)_5(O)]^{3+}$ and back to $[Fe(H_2O)_6]^{3+}$** and this causes the **energy gap between the two sets of d–orbitals to change as well**. Hence, **radiation of different wavelengths from the visible light region were absorbed** and the **colour observed changes from yellow to brown to yellow again**.

[1]

(d) Determination of a value for an enthalpy change of decomposition, ΔH^2_{decomp}, using a graphical method

A student performed the experiment you performed in **1(b)(i)** but measured the temperature at timed intervals. The student added **FA 2**, containing the $[Fe(H_2O)_6]^{3+}$ catalyst, to the **FA 1** at four minutes.

The results from this experiment are plotted on the grid in Fig. 1.1.

(i) Draw two best-fit straight lines, the first taking into account the points before the **FA 2** was added and the second taking into account the points after the reaction had finished.

Extrapolate (extend) both lines to four minutes.

Determine a value for the maximum temperature change, ΔT^2_{max}, at t = 4.0 min using the graph.

$\Delta T^2_{max} = 33.00 - 21.45 = 11.55°\text{ C} = \underline{\mathbf{+11.6\ °C}}$

ΔT^2_{max} = +11.6 °C [2]

(ii) Calculate a second value for the enthalpy change for the decomposition reaction shown in equation **1**, ΔH^2_{decomp}, using this value of ΔT^2_{max}.

Average heat change, $q = -mc\Delta T^2_{max} = -(100 \times 1.00) \times 4.18 \times 11.55 = -4827.9\text{ J} \approx -4.828\text{ kJ}$

$\Delta H = \dfrac{-4.828\text{ kJ}}{0.0425\text{ mol}} = -113.6\text{ kJ mol}^{-1}$ $\left(\text{for } H_2O_2 \rightarrow H_2O + \tfrac{1}{2}O_2\right)$

$\therefore \Delta H^1_{decomp} = 2 \times -113.6 = \underline{\mathbf{-227\text{ kJ mol}^{-1}}}$

ΔH^2_{decomp} = –227 kJ mol^{-1} [1]

(iii) State which of the temperature changes, ΔT^1_{max} or ΔT^2_{max}, is likely to be more accurate. Explain why your chosen value is likely to be the more accurate one.

ΔT^2_{max} is more accurate as it **accounts for heat lost to surrounding by monitoring the temperature over time** and hence, the rate of cooling by surrounding is accounted for. [1]

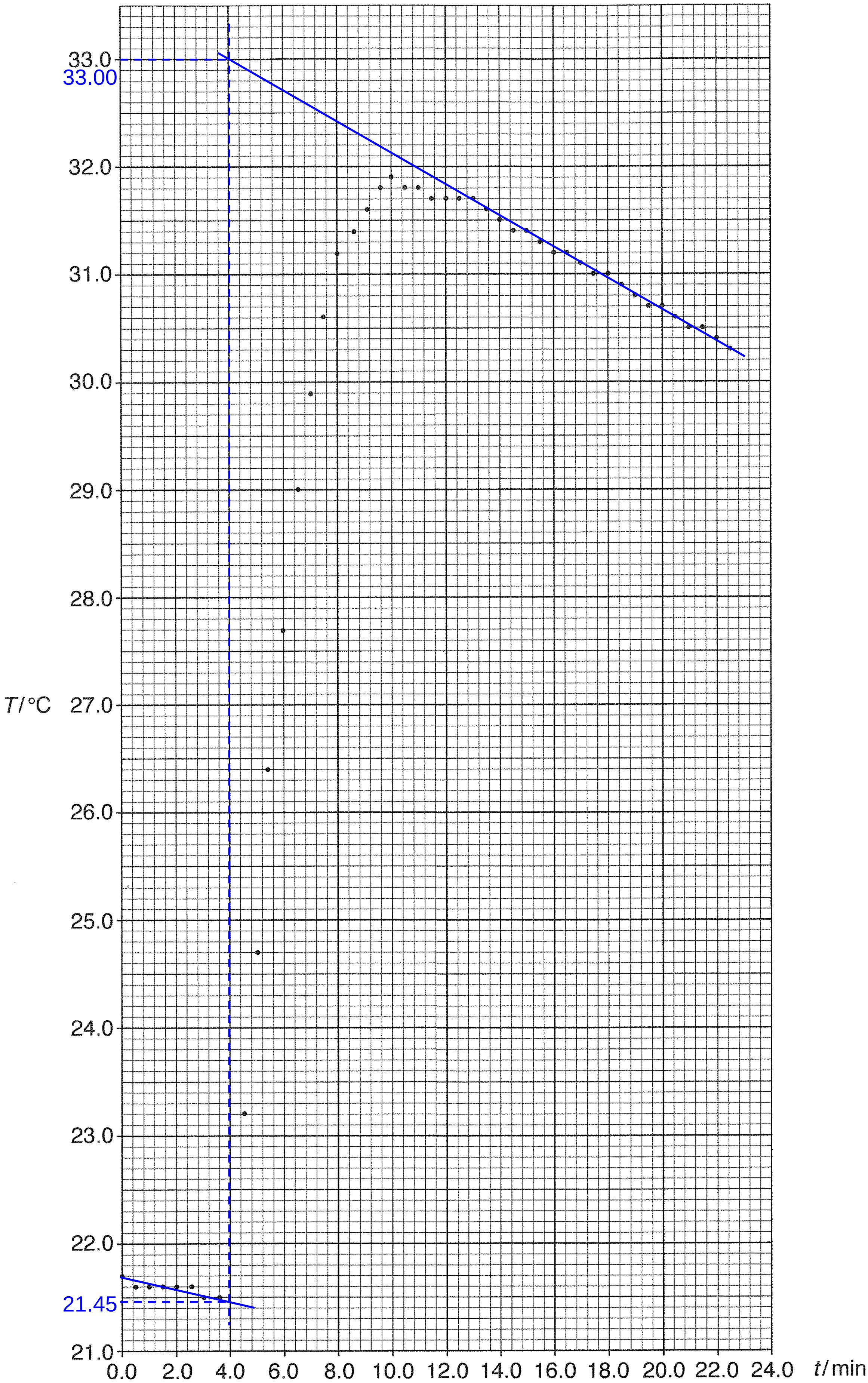

Fig. 1.1

[Total: 17]

2 **Investigation of the kinetics of the catalysed decomposition of hydrogen peroxide**

The concentration of **FA 1** is too high for the **FA 1** to be titrated directly against **FA 3**. A diluted solution of **FA 1** has been provided for you. This diluted solution is **FA 4**.

FA 3 is 0.0200 mol dm^{-3} potassium manganate(VII), $KMnO_4$.
FA 4 is 0.170 mol dm^{-3} aqueous hydrogen peroxide, H_2O_2.

You are also provided with 0.2 mol dm^{-3} sulfuric acid, H_2SO_4.

You will add a measured volume of **FA 2** to a measured volume of **FA 4** and, at timed intervals, you will transfer aliquots (portions) of the reaction mixture.

It is necessary that you titrate each aliquot against **FA 3** **before** transferring the next aliquot.

Acidified $KMnO_4$ and H_2O_2 react as shown in equation **2**.

equation **2** $2MnO_4^-(aq) + 5H_2O_2(aq) + 6H^+(aq) \rightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5O_2(g)$

(a) (i) Preparation and titration of the reaction mixture

Notes: You will perform each titration **once** only. Great care must be taken that you do not overshoot the end-point.

Once you have started the stopwatch, it must continue running for the duration of the experiment. You must **not** stop it until you have finished this experiment.

You should aim to transfer your first aliquot within the first three minutes of starting the reaction.

You should aim **not** to exceed a maximum reaction time of 25 minutes for this experiment.

In an appropriate format in the space provided, prepare a table in which to record for each aliquot
- the time of transfer, t, in minutes and seconds,
- the decimal time, t_d, in minutes, to 0.1 min, for example, if t = 4 min 33 s then t_d = 4 min + 33/60 min = 4.6 min,
- the burette readings and the volume of **FA 3** added.

1. Fill a burette with **FA 3**.
2. Using a measuring cylinder, add 100.0 cm^3 of **FA 4** to the conical flask labelled **reaction mixture**.
3. Using a measuring cylinder, add 2.0 cm^3 of **FA 2** to the same conical flask. Start the stopwatch and swirl the mixture thoroughly to mix its contents.
4. Using a measuring cylinder, add 50.0 cm^3 of 0.2 mol dm^{-3} sulfuric acid to a second conical flask.
5. Transfer a 10.0 cm^3 aliquot (portion) of the reaction mixture to a 10 cm^3 measuring cylinder, using a dropping pipette.
6. **Immediately** transfer this aliquot into the second conical flask and vigorously swirl the mixture. Read and record the time of transfer in minutes and seconds, to the nearest second, when the aliquot is added.
7. **Immediately** titrate the H_2O_2 in the second conical flask with **FA 3**. The end-point is reached when a permanent **pale** pink colour is obtained. Record your titration results.
8. Wash out the second conical flask with water.
9. Repeat steps **4** to **8** until a total of **five** aliquots have been titrated and their results recorded.

Results

Transfer time t	1min 55s	4min 30s	7 min 30s	11min 0s	13min 30s
t_d / min	1.9	4.5	7.5	11.0	13.5
Initial burette reading / cm^3	0.00	0.00	23.50	0.00	14.40
Final burette reading / cm^3	29.40	23.50	42.10	14.40	26.80
Volume of **FA 3** / cm^3	29.40	23.50	18.60	14.40	12.40

[4]

(ii) Plot a graph of the volume of **FA 3** added, on the y-axis, against decimal time, t_d, on the x-axis on the grid in Fig. 2.1.

Draw the most appropriate best-fit curve taking into account all of your plotted points. Extrapolate (extend) this curve to t_d = 0.0 min.

Fig. 2.1

[3]

(b) The **initial rate** of change of the concentration of hydrogen peroxide, **FA 4**, [H_2O_2], can be determined from the gradient of the tangent to the graph in Fig. 2.1 at time t_d = 0.0 min.

(i) Draw a tangent to your graph in Fig. 2.1 at time t_d = 0.0 min.

Determine the gradient of this line, showing clearly how you did this.

$$\text{Gradient} = \frac{35-0}{0-10} = \underline{\mathbf{-3.50\ cm^3\ min^{-1}}}$$

gradient = ...–3.50.......... $cm^3\,min^{-1}$ [3]

(ii) Use your gradient to determine the rate of change of the amount of MnO_4^- ions required in mol min^{-1}.

$$\text{Amount of } MnO_4^- \text{ in } 3.50\ cm^3 = [MnO_4^-] \times \frac{3.50}{1000} = 0.0200 \times \frac{3.50}{1000} = 7.00 \times 10^{-5}\ mol$$

$$\text{Rate of change of amount of } MnO_4^- \text{ required} = \underline{\mathbf{-7.00 \times 10^{-5}\ mol\ min^{-1}}}$$

rate of change of the amount of MnO_4^- ions required = ...-7.00×10^{-5}.......... mol min^{-1} [1]

(iii) Determine the amount of H_2O_2 decomposed per minute and hence deduce the rate of change of [H_2O_2] at t_d = 0.0 min, in mol $dm^{-3}\,min^{-1}$.

$$\text{Amount of } H_2O_2 \text{ decomposed per minute} = \frac{5}{2} \times 7.00 \times 10^{-5} = 1.75 \times 10^{-4}\ mol$$

$$\text{Change in } [H_2O_2] \text{ per minute} = -\frac{1.75 \times 10^{-4}\ mol}{\frac{10}{1000}\ dm^3} = -0.0175\ mol\ dm^{-3}$$

$$\text{Rate of change of } [H_2O_2] = \underline{\mathbf{-0.0175\ mol\ dm^{-3}\ min^{-1}}}$$

rate of change of [H_2O_2] at t_d = 0.0 min = ...–0.0175.......... mol $dm^{-3}\,min^{-1}$ [2]

(iv) It has been claimed that the decomposition of hydrogen peroxide is first order with respect to [H_2O_2].

State whether you agree or disagree with this claim. Use evidence from your graph in Fig. 2.1 to support your answer.

I **disagree**, since the **half-life** for the decomposition of H_2O_2 is **not constant**,

increasing from 8.3 min to 8.7 min to 9.7 min

....

....

....

....

....[2]

(c) Planning

The decomposition of hydrogen peroxide, shown in equation **1**, is also catalysed by the enzyme catalase, present in liver.

The rate equation for this decomposition is shown.

$$\text{rate} = k[H_2O_2][\text{catalase}]$$

If the concentration of the catalase is kept constant, the rate equation becomes

$$\text{rate} = k'[H_2O_2]$$

where $k' = k[\text{catalase}]$.

The activation energy, E_a, and the pre-exponential factor, A, which is a constant, can be determined from the equation.

$$k' = Ae^{\frac{-E_a}{RT}}$$

T is the reaction temperature in kelvin.
k' is the rate constant at a chosen temperature.

The procedure you followed in **2(a)(i)** can be modified to investigate the effect of temperature, T, on the rate of decomposition of H_2O_2. The activation energy, E_a, and the pre-exponential factor, A, can be graphically determined.

Plotting $\ln k'$ against $\frac{1}{T}$ gives a straight line of best fit. The gradient of this line is $\frac{-E_a}{R}$, where R is the molar gas constant.

(i) Plan an investigation, based on the experiment described in **2(a)(i)**, to determine the effect of temperature, T, on the rate of decomposition of H_2O_2 with catalase.

You may assume that you are provided with

- $500\,cm^3$ of $0.170\,mol\,dm^{-3}$ hydrogen peroxide, H_2O_2,
- a piece of liver,
- a scalpel (knife),
- $0.0200\,mol\,dm^{-3}$ potassium manganate(VII), $KMnO_4$,
- $0.2\,mol\,dm^{-3}$ sulfuric acid, H_2SO_4,
- the equipment normally found in a school or college laboratory.

In your plan you should include brief details of

- the reactants and conditions that you would use,
- the apparatus that you would use in addition to that specified in **2(a)(i)**,
- the procedure that you would follow and the measurements that you would take,
- how you would determine the initial rate for each experiment.

Chemicals:

- **FA 3** is 0.0200 mol dm^{-3} potassium manganate(VII), $KMnO_4$
- **FA 4** is 500 cm^3, 0.170 mol dm^{-3} aqueous hydrogen peroxide, H_2O_2
- **FA 5** is 0.2 mol dm^{-3} sulfuric acid, H_2SO_4
- Catalase from a piece of liver

Apparatus:

- Water bath
- Scalpel
- Apparatus used in **2(a)(i)**

Method:

The initial rates of the reaction between H_2O_2 and catalase will be determined from graph(s) of V_{FA3} against time, t. From the comparison of the initial rates for the five specific temperature conditions (25, 35, 45, 55, and 65 °C), I can determine the effect of temperature on the rate of decomposition of H_2O_2 with catalase. This will be done based on the experiment described in **2(a)(i)**, with the addition of keeping temperature constant using a water bath and ensuring that the concentration of H_2O_2 and H_2SO_4 and the volume(mass) of the liver (to ensure constant concentration of catalase) will be kept constant for all five temperatures.

Procedure:

1. Fill a burette with **FA 3**.
2. Prepare the water bath and set the temperature to **25 °C**.
3. Using a 100 cm^3 measuring cylinder, add 100.0 cm^3 of **FA 4** to the conical flask labelled **reaction mixture**. Place this conical flask into the water bath and allow it to equilibrate to the set temperature.
4. Using a 100 cm^3 measuring cylinder, add 50.0 cm^3 of **FA 5** to a second conical flask.
5. Using a scalpel, cut a 1cm × 1 cm × 1cm piece of liver (OR weigh 1.00 g of liver using a mass balance) and add it to the conical flask labelled reaction mixture. Start the stopwatch and swirl the mixture thoroughly to mix its content.

6. Transfer a 10.0 cm^3 aliquot (portion) of the reaction mixture to a 10 cm^3 measuring cylinder, using a dropping pipette.
7. At time = 2.0 min, **immediately** transfer this aliquot into the second conical flask and vigorously swirl the mixture.
8. **Immediately** titrate the H_2O_2 in the second conical flask with **FA 3**. The end-point is reached when a permanent **pale** pink colour is obtained. Record your titration results.
9. Wash out the second conical flask with water.
10. Repeat steps **6** to **9** for t = 5.0, 8.0, 11.0 and 14.0 min respectively, recording their results.
11. Repeat steps **1** to **11**, changing the temperature of the water bath to **35, 45, 55 and 65 °C respectively**.

Analysis:

1. Using the titration results, determine the volume of **FA 3** used at each time interval.
2. Plot graph(s) of $\mathbf{V_{FA\,3}}$ against time for all five temperature condition.
3. By drawing a tangent at t = 0.0 min, determine the initial rates for rate of decomposition between catalase and H_2O_2 for each temperature condition using the gradient of the tangent.
4. By comparing the initial rates for all the experiments, conclude how temperature will affect the rate of reaction.

[7]

(ii) Briefly describe how you would use results obtained from **2(c)(i)** to determine all necessary values in order to plot a graph of $\ln k'$ against $\frac{1}{T}$.

You do **not** need to perform any of the calculations.

In **(c)(i)**, initial rate of the reactions have been determined for each temperature. Since rate = $k'[H_2O_2]$, if the initial rate is used then $[H_2O_2]$ = 0.170 mol dm^{-3}. So k' = initial rate/0.170. Subsequently, the natural logarithm of k' can be calculated to determine $\ln k'$. $1/T$ can be determined by first converting T in the experiment to kelvin, K, and taking the reciprocal of T in K.

[2]

(iii) Sketch the graph you would expect to obtain from **2(c)(ii)** on the axes in Fig. 2.2.

Explain your answer.

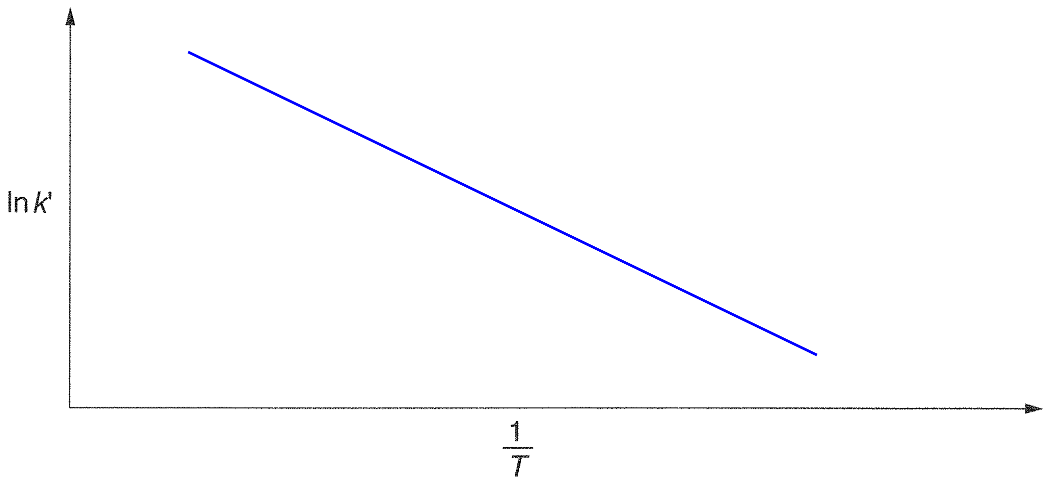

Fig. 2.2

explanation Since $k' = Ae^{-\frac{E_a}{RT}}$, taking natural logarithm on both sides give

$\ln k' = \ln A + \ln e^{-\frac{E_a}{RT}} = \ln A - \frac{E_a}{RT} = -\left(\frac{E_a}{R}\right)\frac{1}{T} + \ln A$, which is of the form $y = mx + c$ if

ln k' is plotted against $1/T$, hence giving a straight line with a negative gradient [2]

(iv) Describe how you would use your graph to determine values for E_a and A.

E_a The gradient of the line will be $-\frac{E_a}{R}$. So E_a will be given by gradient $\times -R$,

where R is the molar gas constant, 8.31 J K^{-} mol^{-1}.

A Extrapolate the line to the y-axis and determine the y-intercept, which will be

ln A. So A will be given by $e^{y\text{-intercept}}$

[3]

[Total: 29]

3 **Investigation of some redox reactions involving transition element ions**

FA 5 is aqueous cobalt(II) chloride, $CoCl_2$.

(a) Perform the tests described in Table 3.1, and record your observations in the table. Test and identify any gases evolved. If there is no observable change, write **no observable change**.

Table 3.1

	tests	observations
1.	Add deionised water to a boiling tube until it is about one quarter full. Add aqueous sodium hydroxide to a second boiling tube until it is about one quarter full. Gently heat the second boiling tube until the liquid just starts to boil. Add one drop of **FA 3** to each boiling tube and shake them thoroughly. Observe the mixtures until no further changes are seen.	In the boiling tube with water, **pink** solution **remains unchanged**. In the boiling tube with NaOH(aq), **purple** solution turns **grey** and then turns **green** upon heating.
While you are waiting, continue with test 2.		
2.	Add ten drops of **FA 5** to a test-tube. Add aqueous ammonia dropwise, with shaking, until no further change is seen. Then add about 1 cm depth of **FA 1** and shake the mixture thoroughly.	**Blue-green ppt partially dissolves** in **excess** NH_3(aq) to give **green solution**.* Upon addition of **FA 1**, the blue-green ppt turns **dark brown** and green solution turns **brown**.

[4]

* Theoretically, a **blue** basic salt, Co(OH)Cl, is first formed when NH_3(aq) is added:

$$Co^{2+} + NH_3 + H_2O + Cl^- \rightarrow Co(OH)Cl + NH_4^+$$

This should dissolve in excess NH_3(aq) to give a **yellowish-brown** solution of $[Co(NH_3)_6]^{2+}$:

$$Co(OH)Cl + 6NH_3 \rightarrow [Co(NH_3)_6]^{2+} + Cl^- + OH^-$$

which slowly turns **brownish-red** is exposed to air:

$$4[Co(NH_3)_6]^{2+} + O_2 + 2H_2O \rightarrow 4[Co(NH_3)_6]^{3+} + 4OH^-$$

Hydrogen peroxide oxidises the complex ion more rapidly:

$$2[Co(NH_3)_6]^{2+} + H_2O_2 \rightarrow 2[Co(NH_3)_6]^{3+} + 2OH^-$$

(b) Consider your observations in Table 3.1.

(i) A small amount of oxygen is produced in one of the boiling tubes used in test 1 in Table 3.1. It is unlikely that you will have noticed this.

Suggest an explanation for the colour changes you observed in test 1 in Table 3.1, in terms of the physical and chemical changes occurring.

Identify the boiling tube in which the oxygen is produced.

The boiling tube with NaOH(aq) is the one in which oxygen is produced.

In the boiling tube with water, **only physical change has occurred** during **dilution of $KMnO_4$. No chemical reaction has occurred.**

In the boiling tube with NaOH(aq), **chemical reaction has occurred** as **purple MnO_4^- oxidises OH^- to O_2, itself reduced to green MnO_4^{2-}. The physical change observed of the grey solution is due to mixture of purple MnO_4^-(aq) and green MnO_4^{2-}(aq).** [2]

(ii) The original colour of **FA 5** is due to the presence of $[Co(H_2O)_6]^{2+}$ ions in the solution.

Identify, from your observations, the cobalt-containing species formed during test 2 in Table 3.1.

- **Pink solution is $[Co(H_2O)_6]^{2+}$**
- **Blue-green ppt is $Co(OH)_2$**
- **Green solution is $[Co(NH_3)_6]^{2+}$**
- **Brown solution is $[Co(NH_3)_6]^{3+}$**

[2]

(iii) Compare the role of hydrogen peroxide in test 2 in Table 3.1 with its role in equation **2**, on page 8. Explain your answer in terms of oxidation state changes.

H_2O_2 is **oxidising agent in test 2** as it oxidises $[Co(NH_3)_6]^{2+}$ to form $[Co(NH_3)_6]^{3+}$ where **oxidation state of Co increases from +2 to +3.**

H_2O_2 is **reducing agent in equation 2** as it reduces MnO_4^-(aq) to form Mn^{2+}(aq) where **oxidation state of Mn decreases from +7 to +2.**

[1]

[Total: 9]